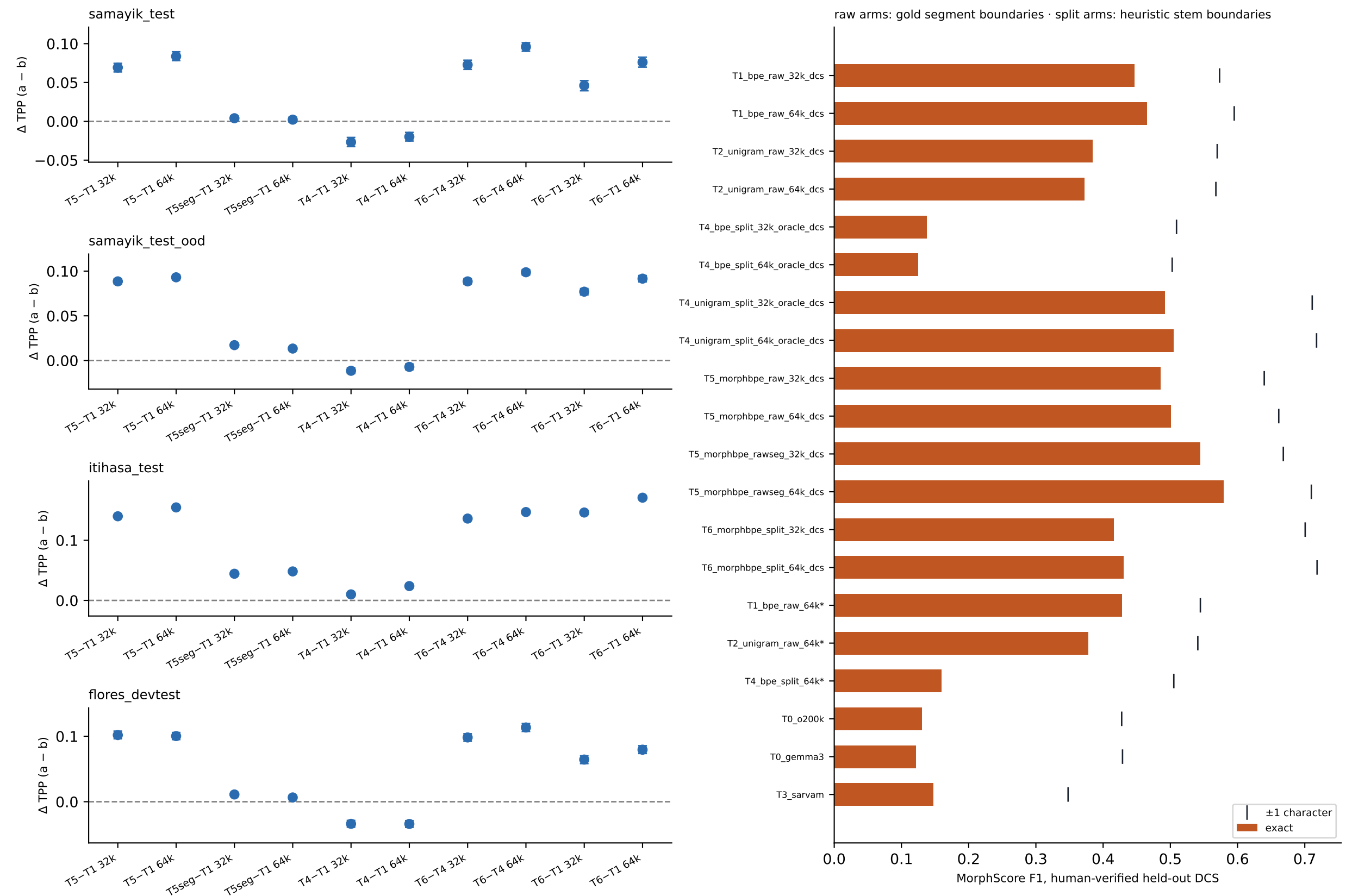


Morpheme-constrained merges: paired TPP deltas and MorphScore
 left: $\text{TPP}(a) - \text{TPP}(b)$ on parallel text, same sentences and same English side, 95% paired bootstrap CI · right: boundary F1 on held-out DCS



All `dcs` arms are trained on the DCS training split; `oracle` arms are trained on DCS's own gold segmentation, which is an upper bound on what splitting can buy, while split arms are *evaluated* on ByT5-reconciled text. `T5` is constrained on gold segment boundaries and heuristic stem boundaries, `T5seg` on the gold segment boundaries alone, `T6` on heuristic stem boundaries inside the gold split. Arms marked provisional (*) are trained on the parallel corpora, not on DCS, and are shown for continuity with Experiments 02-03 only. MorphScore for raw arms uses DCS segment boundaries; for split arms it uses the **heuristic stem** boundary derived from the lemma, which is not annotation, so the two columns are not one series. Thin markers are the ± 1 -character tolerant variant. The English pivot is cross-corpus (E1 was trained on the parallel English side, not on DCS), which is why the deltas — where the English side cancels — carry the verdict and the levels do not.